# Hwanhee Cho

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Generated 2026-10-25 — first-author papers highlighted

Senior researcher in the Electric Railway Power Research Division at the Korea Railroad Research Institute (KRRI). Research focuses on power system stability and oscillation analysis, and on electric railway power supply systems including energy-saving operation, hybrid energy storage, and MVDC.

## Education

- B.S. in Electrical and Electronic Engineering, Dankook University, Yongin, South Korea (2007 – 2014)
- Integrated M.S. and Ph.D. in Electrical and Electronic Engineering, Korea University, Seoul, South Korea (2014 – 2020)

## Experience

- Secondment to the Korea Agency for Infrastructure Technology Advancement (KAIA), Anyang-si, South Korea (Feb. 23, 2026 – Feb. 22, 2027)
- Senior Researcher, Electric Railway Power Research Division, Korea Railroad Research Institute (KRRI), Uiwang-si, South Korea (since 2020)

## Research Interests

- Power Systems: Power System Oscillations, PMU, Power System Stability
- Electric Railway Systems: Energy Saving, Hybrid ESS, MVDC, AC/DC Hybrid, Railway Computation

## Skills

- Engineering tools: PSSE, PSCAD, MATLAB, Simulink, OPAL-RT (eFPGASim, eMEGASim)
- Programming: C, Python

## Publications (31)

- B. Kim, J. Park, S. Oh, H. Cho, B. Lee, “RMS-Based PLL Stability Limit Estimation Using Maximum Phase Error for Power System Planning in Weak Grids,” Energies, Jan. 2026. doi:10.3390/en19010281
- H. Cho, J. Kim, H. Lee, H. Jung, “Development of a TCP/IP-Based Hardware-in-the-Loop Simulation Platform for Virtual Train Operation and Energy Storage System Integration,” The Transactions of the Korean Institute of Electrical Engineers, Dec. 2025. doi:10.5370/kiee.2025.74.12.2330
- B. Kim, J. Park, H. Cho, S. Nam, B. Lee, “An Enhanced Geometric Featured Extraction Method for Real-Time Oscillation Detection in Power System,” IEEE Access, Aug. 2025. doi:10.1109/access.2025.3597375
- M. Song, J. Kim, H. Cho, H. Kim, H. Jung, “Universal MNRV DPWM Methodology for Effective Voltage Balancing in Diode-Clamped Multilevel Topologies,” The Transactions of the Korean Institute of Electrical Engineers, Jul. 2025. doi:10.5370/kiee.2025.74.7.1266
- H. Cho, J. Kim, M. Song, C. Yun, G. Cho, Z. Tian, “Hybrid PI and Fuzzy Logic Control for Energy Optimization in Train Operations,” IEEE Access, Apr. 2025. doi:10.1109/access.2025.3555592
- H. Cho, J. Kim, H. Lee, M. Song, “Research on DC Railway Vehicle for Voltage Stabilization of Electric Railway Substations Using Operation Data-Based Control Strategy,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2024. doi:10.5370/kiee.2024.73.10.1779

- M. Song, J. Kim, H. Cho, “Three-Phase Diode-Clamped Four-Level PWM Inverter With Offset Voltage Injection,” IEEE Access, Jul. 2024. doi:10.1109/access.2024.3424413
- H. Cho, J. Kim, H. Lee, M. Song, “Simultaneous Computation of Traction Power Supply System Using State-Space Nodal Grouping Method,” Journal of Electrical Engineering & Technology, Jan. 2024. doi:10.1007/s42835-023-01716-4
- J. Kim, H. Cho, H. Lee, M. Song, J. Park, H. Jung, “A Study on the Virtual Operation of Energy Storage Devices using the Electric Railway System HILS Platform,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2023. doi:10.5370/kiee.2023.72.10.1291
- H. Noh, H. Cho, S. Choi, B. Lee, “Mitigating Subsynchronous Torsional Interaction Using Geometric Feature Extraction Method,” Sustainability, Dec. 2022. doi:10.3390/su142316110
- J. Kim, H. Cho, H. Jung, H. Kim, “Reliability Demonstration of Auxiliary Power Unit in Railway Vehicles using the Virtual Operation Simulator,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2022. doi:10.5370/kiee.2022.71.10.1460
- J. Kim, J. Ryu, H. Cho, Y. Oh, H. Lee, “A Study on Optimal Operation Strategy of Hydrogen Charging Facilities at Hydrogen Fuel Cell Train Depots,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2022. doi:10.5370/kiee.2022.71.10.1467
- M. Song, H. Jung, H. Kim, J. Kim, G. Cho, H. Cho, “Pulse-Amplitude-Modulation Full-Bridge Diode-Clamped Three-Level LLC Resonant Converter With Offset Voltage Injection Method,” The Transactions of the Korean Institute of Electrical Engineers, Sep. 2022. doi:10.5370/kiee.2022.71.9.1342
- H. Cho, J. Kim, H. Jung, H. Kim, “Simultaneous DC Railway Power System Analysis Method Using Model-Based TPS,” Applied Sciences, Jul. 2022. doi:10.3390/app12146929
- J. Kim, H. Cho, J. Ryu, H. Kim, “Development and applicability test of fast charge/discharge system for DC 1500V railway vehicles,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2021. doi:10.5370/kiee.2021.70.10.1559
- H. Cho, J. Kim, H. Kim, “Time-series based System Stability Estimation for Railroad Vehicle PMSM Using Real-time Simulator,” The Transactions of the Korean Institute of Electrical Engineers, Oct. 2021. doi:10.5370/kiee.2021.70.10.1526
- H. Cho, N. Choi, S. Nam, B. Lee, “Extended Geometric Feature Extraction Process for Detecting Multiple Frequency Oscillations in KEPCO System,” IEEE Access, Jun. 2021. doi:10.1109/access.2021.3090360
- H. Noh, H. Cho, S. Lee, B. Lee, “STATCOM with SSR damping controller using geometric extraction on phase space reconstruction method,” International Journal of Electrical Power & Energy Systems, Sep. 2020. doi:10.1016/j.ijepes.2020.106017
- D. Kim, H. Cho, B. Park, B. Lee, “Evaluating Influence of Inverter-based Resources on System Strength Considering Inverter Interaction Level,” Sustainability, Apr. 2020. doi:10.3390/su12083469
- J. Jung, H. Cho, B. Park, S. Nam, K. Hur, B. Lee, “Enhancement of linearity and constancy of PMU-based voltage stability index: application to a Korean wide-area monitoring system,” IET Generation, Transmission & Distribution, Feb. 2020. doi:10.1049/iet-gtd.2019.0773
- H. Cho, N. Choi, B. Lee, “Oscillation Recognition Using a Geometric Feature Extraction Process Based on Periodic Time-Series Approximation,” IEEE Access, Feb. 2020. doi:10.1109/access.2020.2974259
- H. Noh, H. Cho, B. Lee, S. Lee, “Toward Mitigating Power System Oscillation : STATCOM Controller With Delay Reconstruction Algorithm,” 2019 IEEE 8th International Conference on Advanced Power System Automation and Protection (APAP), Oct. 2019. doi:10.1109/apap47170.2019.9225023
- N. Choi, H. Cho, B. Lee, “Development of Floquet Multiplier Estimator to Determine Nonlinear Oscillatory Behavior in Power System Data Measurement,” Energies, May 2019. doi:10.3390/en12101824
- N. Choi, B. Park, H. Cho, B. Lee, “Impact of Momentary Cessation Voltage Level in Inverter-Based Resources on Increasing the Short Circuit Current,” Sustainability, Feb. 2019. doi:10.3390/su11041153
- S. Oh, H. Shin, H. Cho, B. Lee, “Transient Impact Analysis of High Renewable Energy Sources Penetration According to the Future Korean Power Grid Scenario,” Sustainability, Nov. 2018. doi:10.3390/su10114140

- H. Cho, S. Oh, S. Nam, B. Lee, “Non-linear dynamics based sub-synchronous resonance index by using power system measurement data,” IET Generation, Transmission & Distribution, Aug. 2018. doi:10.1049/iet-gtd.2018.5414
- J. Jeong, M. Bae, H. Cho, B. Lee, H. Shim, J. Lee, “Empirical approach to regulate bus voltage range in power systems,” Journal of International Council on Electrical Engineering, Jan. 2015. doi:10.1080/22348972.2015.1110884
- H. Cho, M. Bae, J. Jeong, B. Lee, “Reactive power compensation using OPF for various load conditions,” Proceedings of the KIEE Conference, 2015.
- M. Son, H. Cho, Y. Sohn, “Magnetic fabric (anisotropy of magnetic susceptibility) constraints on emplacement mechanism of clastic dikes: an example from the Cretaceous Dadaepo Basin in SE Korea,” AGUFM, Dec. 2014.
- S. Han, Y. Song, F. Cho, S. Ryu, G. Cho, Y. Song, H. Cho, “Magnetic field anisotropy based MR tractography,” Journal of Magnetic Resonance, Aug. 2011. doi:10.1016/j.jmr.2011.07.021
- T. Zhao, C. Hou, H. Fujiwara, H. Cho, J. Harrell, A. Khapikov, “Study of 360° domain walls in NiFe/NiO film by tip–sample interaction on magnetic force microscope,” Journal of Applied Physics, May 2000. doi:10.1063/1.372745